# THE FOURIER ENTROPY OF LINEAR MONOTONE DNFS ON FEW VARIABLES

YITZCHAK SHMALO

Abstract. A monotone formula in disjunctive normal form (DNF) is linear if any two of its terms share at most one variable. Over all linear monotone DNFs with at most eight relevant variables, of which there are 552,251 isomorphism classes on exactly eight, we determine by exhaustive enumeration with certified logarithms the largest value of the ratio $H(f)/I(f)$ of Fourier entropy to total influence and the largest value of $\min_i E_i(f)/\mathrm{Inf}_i(f)$, where $E_i$ is the erasure entropy at $i$. They are $3.8891054198\ldots$ and $3.3047013224\ldots$, and each is attained by a single class. A transversal design on 21 variables has $H/I = 4.1346\ldots$, so neither value is the supremum over the class. We then give three sufficient conditions for a coordinate with $E_i \le c\,\mathrm{Inf}_i$, $c$ an absolute constant, in a linear monotone DNF: a load certificate that assumes no bound on widths or degrees, a criterion that involves the loads of the variables of a single term only, and a bound in terms of the bias of $f$. On the affine transversal designs with $k + 2\lceil \log_2 k \rceil + 1$ groups of $2^k$ points each, all of these conditions fail at every coordinate, as do the width and variance bounds of earlier work, while $\Pr[F = 0]$ tends to zero more slowly than any exponential in the number of variables.

## 1. Introduction

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ with $\chi_S(x) = (-1)^{\sum_{j\in S} x_j}$, and let $U$ be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy is $H(f) = H(U)$, the influence of $i$ is $\mathrm{Inf}_i(f) = \Pr[i \in U]$, and $I(f) = \sum_i \mathrm{Inf}_i(f)$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [4] asks for an absolute $C$ with $H(f) \le C\, I(f)$ for every Boolean $f$. It is known for symmetric functions [16] and for read-once formulas, over gates of bounded arity [15] and over AND and OR gates [3]. It is also known for read-$k$ decision trees with $k$ fixed [23] and for linear threshold functions [21]. Its min-entropy version is known for read-$k$ DNFs with $k$ fixed [17, 1], and Kelman, Kindler, Lifshitz, Minzer and Safra [9] show that the spectrum of a Boolean function of constant variance is concentrated on $2^{O(I(f)\log I(f))}$ characters, a weaker form of the conjecture. For every Boolean $f$, Han [5] proved

$$H(f) \le \Big(2 + \frac{3}{\ln 2}\Big) I(f) + \sum_i \mathrm{Inf}_i(f) \log_2 \frac{1}{\mathrm{Inf}_i(f)},$$

and Li and Han [11] lowered the constant in front of $I(f)$ to the sharp value $1/\ln 2$. The erasure entropy (in the sense of Verdú and Weissman [22]) of $f$ at $i$ is

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big),$$

and the FEI conjecture holds if and only if there is an absolute $c$ such that every non-constant Boolean function has a coordinate with $E_i(f) \le c\,\mathrm{Inf}_i(f)$ [18].

Let $F$ be a monotone DNF and $f = 1 - 2F$. We call $F$ *linear* if any two of its terms share at most one variable. Read-once DNFs are the linear DNFs in which every variable lies in one term. For linear DNFs whose terms all have width $w$ and in which every variable lies in at most

2020 *Mathematics Subject Classification.* 06E30, 42C10, 94A17.

*Key words and phrases.* Fourier entropy-influence conjecture, Boolean functions, Fourier entropy, total influence.

$\Delta \le 2^{(1-\nu)(w-1)}$ terms, $E_i/\mathrm{Inf}_i = 2 + o(1)$ at every coordinate as $w \to \infty$ [20], and a DNF of width $w$ has a coordinate with $E_i/\mathrm{Inf}_i \le w + 3 + \log_2 e$ [19]; for a monotone DNF one term of width $w$ is enough (Section 2). The bound of [19] grows with $w$, so it does not give a uniform bound when the widths are unbounded. This paper concerns linear monotone DNFs with no restriction on widths or degrees. Since the minimal DNF of a monotone function is unique, linearity is a property of the function.

1.1. **Small dimensions.** Call two functions equivalent if they differ by a permutation of the variables; $H$, $I$ and the multiset of pairs $(E_i, \mathrm{Inf}_i)$ are invariants.

**Theorem 1.** *For $n \le 8$, Table 1 lists the number of equivalence classes of linear monotone functions with exactly $n$ relevant variables, the number of such functions on a fixed set of $n$ variables, and the largest values of $H/I$ and of $\min_i E_i/\mathrm{Inf}_i$ over them. Over all linear monotone functions with at most eight relevant variables,*

$$3.30470132244542 \le \max \min_i \frac{E_i(f)}{\mathrm{Inf}_i(f)} \le 3.30470132245649,$$

*attained only by $F = x_1x_2 \vee x_2x_3 \vee x_3x_4 \vee x_4x_1 \vee x_5x_6 \vee x_6x_7 \vee x_7x_8 \vee x_8x_5$, and*

$$3.88910541981418 \le \max \frac{H(f)}{I(f)} \le 3.88910541981709,$$

*attained only by $F = x_1 \vee x_2 \vee x_3 \vee x_4 \vee x_5x_6 \vee x_6x_7 \vee x_7x_8 \vee x_8x_5$.*

| $n$ | classes | functions | $\max \min_i E_i/\mathrm{Inf}_i$ | $\max H/I$ |
|---|---|---|---|---|
| 2 | 2 | 2 | 2.0000000000 | 2.0000000000 |
| 3 | 5 | 9 | 2.6724889840 | 2.9558895823 |
| 4 | 16 | 97 | 2.6340033909 | 3.4024755512 |
| 5 | 72 | 2,625 | 2.6958875202 | 3.6237403997 |
| 6 | 592 | 185,521 | 2.9444953039 | 3.7397640179 |
| 7 | 10,808 | 35,406,319 | 2.9546838142 | 3.8043570698 |
| 8 | 552,251 | 18,785,639,553 | 3.3047013225 | 3.8891054198 |

TABLE 1. Linear monotone functions with exactly $n$ relevant variables. Values rounded to ten decimals; at $n = 8$ the second largest values are 3.1084496901 and 3.8663646986.

As far as we know this is the first exhaustive computation of the FEI ratio on a class of this size. Biswas and Sarkar [2] searched all Boolean functions on at most five variables for the min-entropy version of the conjecture; the largest ratio of min-entropy to influence they found there is 16/7.

The maxima of Theorem 1 are not suprema over the class. The function $\mathrm{AND}_n$ is a linear DNF with one term, and $H/I \to 4$ along it. The transversal designs of Section 7, which have many intersecting terms of equal width, give larger values: on 21 variables, with 49 terms of width 3 and every variable in 7 of them, $H/I = 4.1345862211$ (Table 2). On these designs $\min_i E_i/\mathrm{Inf}_i$ stays between 2.10 and 2.72. All these values lie well below the best known lower bound $C > 6.5218$ on the constant of the conjecture, from an explicit balanced monotone function on 18 variables [14], so they say nothing about $C$ itself. Earlier lower bounds are due to O'Donnell and Tan [15], Hod [10] and Mosaic Intelligence [12, 13].

1.2. **Certificates and an obstruction.** For a variable $j$ put $\lambda_j = \sum_{t \ni j} 2^{-(|t|-1)}$, the sum over the terms through $j$; we call it the *load* of $j$. In a term of width $L+1$ that meets no other term every variable has load $2^{-L}$.

**Theorem 2.** *Let $F$ be a linear monotone DNF and $t$ a term of width $L+1$ with $L \geq 3$. If every variable $j$ of $t$ satisfies*

$$\lambda_j \leq \Big(1 + \frac{1}{64L^2}\Big)2^{-L},$$

*then $E_i(f) < 6\,\mathrm{Inf}_i(f)$ for every $i \in t$.*

Nothing is assumed about the variables outside $t$, and $t$ need not be a shortest term. Theorem 2 is proved by a variant of the argument for a finite certificate, Proposition 6, which bounds $|E_i/\mathrm{Inf}_i - 2|$ by a quantity computed from the terms through $i$ and the loads of their variables, with no maximum width or maximum degree. The third condition is a bound by the bias of $f$ (Lemma 10): if $m = \min(\Pr[f=1], \Pr[f=-1]) > 0$ then $H(f)/I(f) < 2 + 2n/\log_2(1/m)$, for every Boolean $f$ on $n$ variables.

Section 7 shows that these conditions do not combine into a proof of the FEI conjecture for linear monotone DNFs. For $Q$ a prime power and $w \leq Q$ let $\mathrm{TD}(w, Q)$ be the DNF whose variables are the points of $X \times \mathbb{F}_Q$ for a $w$-set $X \subseteq \mathbb{F}_Q$ and whose terms are the graphs of the affine maps $\xi \mapsto \sigma\xi + \tau$ restricted to $X$.

**Theorem 3.** *Let $Q = 2^k$ and $w = k + 2\lceil \log_2 k \rceil + 1$, and let $F_k = \mathrm{TD}(w, Q)$. As $k \to \infty$:*

1. *every term has width $w \to \infty$, and the hypothesis of Theorem 2 fails for every term;*
2. *the right side of the certificate of Proposition 6 tends to infinity at every coordinate, uniformly over all choices of its parameters;*
3. *$\Pr[F_k = 0] \to 0$ but $\log_2(1/\Pr[F_k = 0])/n \to 0$, so the bias bound tends to infinity;*
4. *the variance bound of Lemma 5 tends to infinity at every coordinate.*

Theorem 3 says nothing about the actual ratios on $F_k$, which we can compute only for small designs; there they are small (Table 2).

## 2. Preliminaries

We use the notation of [20]. Fix $i$ and write $f^{(0)}$ and $f^{(1)}$ for the restrictions $x_i = 0$ and $x_i = 1$. Put $g = \frac{1}{2}(f^{(0)} + f^{(1)})$ and $h = \frac{1}{2}(f^{(0)} - f^{(1)})$, so that $\hat{g}(T) = \hat{f}(T)$ and $\hat{h}(T) = \hat{f}(T \cup \{i\})$ for $T \subseteq [n] \setminus \{i\}$, and $\mathbb{E}h^2 = \mathrm{Inf}_i(f) =: \delta$. With $p(T) = \hat{g}(T)^2$, $q(T) = \hat{h}(T)^2$ and

$$\Phi(a,b) = \eta(a) + \eta(b) - \eta(a+b), \qquad \eta(x) = -x\log_2 x,$$

we have $E_i(f) = \sum_T \Phi(p(T), q(T))$. The function $\Phi$ is concave and homogeneous of degree one, hence superadditive, it is increasing in each argument, and

$$\Phi(a,b) \leq b \log_2\Big(e\,\frac{a+b}{b}\Big). \tag{1}$$

If $F$ is monotone and $f = 1 - 2F$, let $N$ and $M$ be the indicators that $F$ vanishes at $x_i = 0$ and at $x_i = 1$. Then $h = N - M$ is the indicator that $i$ is pivotal, $\hat{h}(\emptyset) = \delta$, and $\hat{g}(T) = 2\hat{N}(T) - \hat{h}(T)$ for $T \neq \emptyset$. We use the separation inequality of [20].

**Proposition 4** ([20, Proposition 2])**.** *Let $F$ be monotone, $f = 1 - 2F$, $\delta = \mathrm{Inf}_i(f) > 0$. Let $\Omega$ be a family of non-empty subsets of $[n] \setminus \{i\}$ and let $\varepsilon \in (0,1]$ and $\kappa \geq 0$ satisfy $\sum_{T \notin \Omega} \hat{h}(T)^2 \leq \varepsilon\delta$ and $\sum_{T \in \Omega} \hat{N}(T)^2 \leq \kappa\delta$. Then*

$$2 - 2\varepsilon - \frac{8}{\ln 2}\sqrt{\kappa} \;\leq\; \frac{E_i(f)}{\delta} \;\leq\; 2 + \frac{8}{\ln 2}\sqrt{\kappa} + \varepsilon\log_2\Big(e\Big[3 + \frac{8\mathrm{Var}N}{\varepsilon\delta}\Big]\Big) + \delta\log_2\big(e[1+\delta^{-2}]\big).$$

The next bound improves the variance lemma of [19] for monotone functions.

**Lemma 5.** *Let $F$ be monotone, $f = 1 - 2F$, $v = \mathrm{Var} f$ and $\delta = \mathrm{Inf}_i(f) > 0$. Then*

$$\frac{E_i(f)}{\delta} \le 2|\mathbb{E} f| + \log_2 \frac{e\, v}{\delta}.$$

*Proof.* First, $\Phi(a,b) \le 2\sqrt{ab}$. By symmetry and homogeneity it suffices to take $b = 1$ and $a = x^2$ with $x \ge 1$. Put $\varphi(x) = 2x \ln 2 - (x^2+1)\ln(x^2+1) + x^2 \ln(x^2)$, which is $\ln 2$ times $2x - \Phi(x^2, 1)$. Then $\varphi(1) = 0$ and $\varphi'(x) = 2\ln 2 - 2x \ln(1 + x^{-2})$. The function $x \ln(1 + x^{-2})$ decreases on $[1, \infty)$: its derivative is $\ln(1+y) - 2y/(1+y)$ with $y = x^{-2} \in (0, 1]$, which is non-positive because it vanishes at $y = 0$ and has derivative $-(1-y)/(1+y)^2 \le 0$. So $\varphi' \ge \varphi'(1) = 0$ and $\varphi \ge 0$.

At $T = \emptyset$, $p = (\mathbb{E} f)^2$ and $q = \delta^2$, so the term is at most $2|\mathbb{E} f|\delta$. By Parseval the non-empty sets carry $p_+ = \sum_{T \ne \emptyset} p(T) = v - \delta$ and $q_+ = \sum_{T \ne \emptyset} q(T) = \delta - \delta^2$. By superadditivity and (1),

$$\sum_{T \ne \emptyset} \Phi(p, q) \le \Phi(p_+, q_+) \le q_+ \log_2 \frac{e(v - \delta^2)}{q_+} \le q_+ \log_2 \frac{ev}{q_+} \le \delta \log_2 \frac{ev}{\delta},$$

since $x \mapsto x \log_2(ev/x)$ increases on $(0, v]$ and $q_+ \le \delta \le v$; here $\delta \le v$ because $\delta \le 2m \le 4m(1-m) = v$ with $m = \min(\Pr[f = 1], \Pr[f = -1]) \le 1/2$. If $\delta = 1$ the non-empty part vanishes. □

For comparison, [19] has $2 + \log_2 e + \log_2(1 + v/\delta)$ for every Boolean $f$. The proof of the width bound of [19] uses a single term: if $t$ is any term of width $w$ of a monotone DNF, then some $i \in t$ has $\mathrm{Inf}_i(f) \ge 2\Pr[F = 0]/(2^w - 1)$, and Lemma 5 gives $E_i/\mathrm{Inf}_i \le 2 + \log_2(2e(2^w - 1))$ there.

## 3. The census

3.1. **Enumeration.** A linear monotone function is determined by its minimal DNF, a linear antichain of subsets of the variables, and the function has exactly $n$ relevant variables when the union of the terms is $[n]$. Singleton terms share no variable with any other term. It therefore suffices to enumerate, up to isomorphism, the linear hypergraphs on $[n]$ with edges of size at least two and isolated vertices allowed, and to add a singleton edge at every isolated vertex.

The enumeration processes widths from $n$ down to 2. At width $w$ it repeatedly adds to each family every $w$-set that meets each existing edge in at most one vertex, keeping one family per isomorphism class, until no new class appears, and then passes to $w - 1$. Every linear hypergraph is reached: list its edges by decreasing width, and its prefixes are linear hypergraphs reached at the earlier stages. Isomorphism is decided by an exact canonical form. The vertices are partitioned by the multiset of the sizes of their incident edges, the partition is refined, and unresolved cells are individualised; at each leaf the relabelled edge list is sorted, and the least leaf is the canonical form. Two families with the same canonical form are related by the permutation read off the two leaves.

3.2. **Completeness.** Two independent counts confirm that no class is missing or repeated. The number of labelled linear hypergraphs on $[n]$ with all edges of size at least two, isolated vertices allowed, is computed by a recursion on pairs: choose an uncovered pair $\{j, j'\}$; either no edge of size at least three contains it, and then $\{j, j'\}$ is either an edge or not, or exactly one such edge does, and its pairs are removed. This is compared with $\sum_G n!/|\mathrm{Aut}(G)|$ over the hypergraphs $G$ of the census, with $|\mathrm{Aut}(G)|$ counted at the canonical leaves. Adding a singleton term at each isolated vertex is a bijection onto the functions with exactly $n$ relevant variables, so both counts are the third column of Table 1, and they agree for every $n \le 8$. The census for $n \le 5$ was also compared with a brute-force enumeration of all monotone functions.

A second program, written separately and sharing no code with the first, reads the census and checks for each family that the terms are distinct, pairwise meet in at most one variable, form an antichain and cover $[n]$; it computes a canonical form by a different method (the least sorted relabelled edge list over all relabellings that preserve the vertex invariant) and the automorphism count, and recomputes both ratios. It finds no invalid family and no repeated class, reproduces the third column of Table 1, and agrees with the certified values below to the precision of double arithmetic.

3.3. **Certified values.** For each function the Walsh transform of $f = 1 - 2F$ is computed in integers, so every $p(T)$ and $q(T)$ is an exact dyadic rational with denominator $4^n$, and $H$, $I$ and every $E_i$ are integer combinations of $\log_2 K$ for integers $K \le 4^n$, divided by $4^n$. Enclosures of $\log_2 K$ are computed in integer arithmetic from $\ln x = 2\sum_{j\ge 0} z^{2j+1}/(2j+1)$ with $z = (x-1)/(x+1) \le 1/3$ and a geometric bound on the tail, rounded outwards to units of $2^{-52}$; sums are accumulated in signed 128-bit integers and every floating conversion and division is followed by an outward step. This encloses $H/I$ and every $E_i/\mathrm{Inf}_i$ for every function in the census, and the intervals in Theorem 1 are the resulting enclosures for the two maximisers. In both cases the upper end of every other class's enclosure lies below the lower end of the maximiser's, which proves uniqueness. The exact values are also available as rational combinations of logarithms of primes.

3.4. **The maximisers.** The maximiser of $\min_i E_i/\mathrm{Inf}_i$ is the OR of two copies, on disjoint variables, of the cycle DNF $\mathrm{CYC}_4 = x_1x_2 \vee x_2x_3 \vee x_3x_4 \vee x_4x_1$; this OR has $H/I = 3.6125175459$, against $2.2169171867$ for $\mathrm{CYC}_4$ alone. The maximiser of $H/I$ is the OR of $\mathrm{OR}_4$ and $\mathrm{CYC}_4$ on disjoint variables. Both maxima grow at every step from $n = 4$ to $n = 8$, so the census gives no indication of the suprema.

## 4. The load certificate

Let $F$ be a linear monotone DNF and $i$ a variable that lies in no singleton term. Let $t$ range over the terms through $i$, put $B_t = t \setminus \{i\}$, $L_t = |B_t| \ge 1$, $a_t = 2^{-L_t}$, so that $\lambda_i = \sum_t a_t$. By linearity the sets $B_t$ are pairwise disjoint. A *background term* is a term not containing $i$. For a variable $j \ne i$ put

$$\ell_j = \sum_{s \ni j,\ i \notin s} 2^{-(|s|-1)}, \qquad \mu_t = \sum_{j \in B_t} \ell_j, \qquad \theta = 1 - \frac{1}{\lambda_i}\Big[\sum_t a_t\mu_t + \sum_{t'<t} a_{t'}a_t\Big].$$

For $j \in B_t$, a term through $j$ other than $t$ does not contain $i$, since it would share two variables with $t$; so $\ell_j = \lambda_j - a_t$. For integers $1 \le k_t \le L_t$ put

$$\gamma_t = a_t \sum_{r<k_t} \binom{L_t}{r}, \qquad \alpha_t = 2^{\binom{k_t}{2}-2k_t} \sum_{J \subseteq B_t,\ |J|=k_t} \frac{\prod_{j\in J}\ell_j}{1 - \frac{1}{2}\sum_{j\in J}\ell_j},$$

and

$$\varepsilon = \min\Big(1, \sum_t \gamma_t\Big), \qquad \kappa = \frac{\sum_t \alpha_t}{\lambda_i\theta}.$$

**Proposition 6.** *If $\theta > 0$, every denominator in $\alpha_t$ is positive, and $\lambda_i \le 1/4$, then*

$$\Big|\frac{E_i(f)}{\mathrm{Inf}_i(f)} - 2\Big| \le 2\varepsilon + \frac{8}{\ln 2}\sqrt{\kappa} + \varepsilon\log_2\Big(e\Big[3 + \frac{8}{\varepsilon\lambda_i\theta}\Big]\Big) + \lambda_i\log_2\big(e[1+\lambda_i{}^{-2}]\big).$$

When all terms have width $w$ and all loads are bounded through the degree, this is [20, Theorem 6]; the proof follows the three lemmas there, and we indicate the changes. Let $N$ be the indicator that every background term fails, $A = \mathbb{E}N$, $A_t$ the indicator that all of $B_t$ equal 1, and $h = N \cdot \mathbf{1}[\bigvee_t A_t]$, so that $\delta = \mathbb{E}h = \mathrm{Inf}_i(f)$.

**Lemma 7.** *$A\lambda_i\theta \le \delta \le A\lambda_i$.*

*Proof.* The upper bound is Harris's inequality [7] for the decreasing $N$ and the increasing $A_t$, and the union bound. For the lower bound fix $t$, let $R_t$ indicate that every background term disjoint from $B_t$ fails, and let $V_t$ be the OR of the residual conjunctions $s \setminus B_t$ of the background terms $s$ meeting $B_t$. Each such $s$ meets $B_t$ in one variable, so $\Pr[V_t] \le \mu_t$, and $R_t$ and $V_t$ do not depend on $B_t$. By Harris, $\Pr[N, A_t] = a_t \Pr[R_t, \overline{V_t}] \ge a_t \Pr[R_t](1-\mu_t) \ge a_t A(1-\mu_t)$; if $\mu_t > 1$ this holds trivially. For $t' \ne t$, $\Pr[N, A_{t'}, A_t] \le A a_{t'} a_t$, since $B_{t'}$ and $B_t$ are disjoint. Bonferroni's inequality gives $\delta \ge A\sum_t a_t(1-\mu_t) - A\sum_{t'<t} a_{t'}a_t = A\lambda_i\theta$. □

Let $\Omega$ be the family of sets $T \subseteq [n] \setminus \{i\}$ with $|T \cap B_t| \ge k_t$ for some $t$.

**Lemma 8.** *$\sum_{T\notin\Omega} \hat{h}(T)^2 \le \varepsilon\delta$.*

*Proof.* Order the terms through $i$ and put $h_t = NA_t \prod_{t'<t}(1-A_{t'})$. These have disjoint supports and sum to $h$, so $\sum_t \|h_t\|_2^2 = \delta$. As in [20, Lemma 4], $h_t = A_t\Psi_t$ with $\Psi_t$ independent of $B_t$, so the weight of $\hat{h}_t^2$ on sets meeting $B_t$ in exactly $r$ variables is $\binom{L_t}{r} a_t \|h_t\|_2^2$, and the weight of $\hat{h}_t^2$ off $\Omega$ is at most $\gamma_t\|h_t\|_2^2$. By the triangle inequality and Cauchy–Schwarz,

$$\Big(\sum_{T\notin\Omega} \hat{h}(T)^2\Big)^{1/2} \le \sum_t \gamma_t^{1/2}\|h_t\|_2 \le \Big(\sum_t \gamma_t\Big)^{1/2} \delta^{1/2}.$$

The bound by $\delta$ is trivial. □

**Lemma 9.** *$\sum_{T\in\Omega} \hat{N}(T)^2 \le A\sum_t \alpha_t$.*

*Proof.* Every $T \in \Omega$ contains a $k_t$-subset $J$ of some $B_t$, and $\sum_{T\supseteq J} \hat{N}(T)^2 = \|D_J N\|_2^2$. Fix such $J$. A background term meets $J$ in at most one variable. Let $R_J$ indicate that every background term avoiding $J$ fails, and $Y_j$, for $j \in J$, the OR of the residual conjunctions $s \setminus \{j\}$ of the background terms $s \ni j$; then $\Pr[Y_j] \le \ell_j$, and as in [20, Lemma 5], $\|D_J N\|_2^2 = 4^{-|J|}\Pr[R_J, \bigwedge_j Y_j] \le 4^{-|J|}\Pr[R_J]\Pr[\bigwedge_j Y_j]$. For $\bigwedge_j Y_j$ a background term $s_j \ni j$ must have its residual conjunction satisfied for each $j$. The $s_j$ are distinct, two of the residual sets share at most one variable, so their union has at least $\sum_j(|s_j|-1) - \binom{|J|}{2}$ variables, and summing over the choices gives $\Pr[\bigwedge_j Y_j] \le 2^{\binom{|J|}{2}} \prod_{j\in J}\ell_j$. Finally $A \ge \Pr[R_J] - \sum_j \frac{1}{2}\Pr[R_J, Y_j] \ge \Pr[R_J](1 - \frac{1}{2}\sum_j \ell_j)$ by Harris, since $x_j$ is independent of $R_J$ and $Y_j$. Hence $\|D_J N\|_2^2$ is at most $A$ times the summand of $\alpha_t$ indexed by $J$; sum over $J$ and $t$. □

*Proof of Proposition 6.* By Lemmas 9 and 7, $\sum_\Omega \hat{N}^2 \le A\sum_t \alpha_t \le \kappa\delta$, and by Lemma 8 the first hypothesis of Proposition 4 holds with $\varepsilon$; note $\varepsilon \ge \lambda_i > 0$ because $\gamma_t \ge a_t$. The sets in $\Omega$ are non-empty. In the upper bound of Proposition 4, $\mathrm{Var}N \le A \le \delta/(\lambda_i\theta)$, and $\delta \le A\lambda_i \le \lambda_i \le 1/4$; the function $x\log_2(e[1+x^{-2}])$ is increasing on $(0, 1/4]$ [20]. The lower bound of Proposition 4 is within the stated error. □

## 5. A criterion on one term

*Proof of Theorem 2.* Put $a = 2^{-L}$, $u = 1/(64L^2)$, $\theta_0 = 1 - Lua$, and fix $i \in t$. The terms through $i$ other than $t$ have total weight $\lambda_i - a \le ua$, and every $j \in t \setminus \{i\}$ has $\ell_j = \lambda_j - a \le ua$. Let $N$,

$A$, $A_t$ and $h$ be as in Section 4, and split $h = h_t + h'$ with $h_t = NA_t$, $h' = N(1-A_t)\mathbf{1}[\bigvee_{t'\neq t} A_{t'}]$, which have disjoint supports. By Harris and the proof of Lemma 7,

$$\|h_t\|_2^2 \le Aa, \qquad \|h'\|_2^2 \le Aua, \qquad Aa\theta_0 \le \delta \le A(1+u)a.$$

Let $\Omega$ be the family of sets that contain at least one variable of $t \setminus \{i\}$. The weight of $\hat{h}_t^2$ off $\Omega$ is $a\|h_t\|_2^2 \le Aa^2$, so

$$\Big(\sum_{T\notin\Omega} \hat{h}(T)^2\Big)^{1/2} \le \sqrt{Aa}\,(\sqrt{a}+\sqrt{u}), \qquad \sum_{T\notin\Omega} \hat{h}(T)^2 \le \varepsilon_0\delta, \quad \varepsilon_0 = \frac{(\sqrt{a}+\sqrt{u})^2}{\theta_0}.$$

The proof of Lemma 9 with $k_t = 1$ for the single block $t \setminus \{i\}$ gives $\sum_\Omega \hat{N}^2 \le A\,Lua/(4(1-ua/2)) \le \kappa_0\delta$ with $\kappa_0 = Lu/(4\theta_0(1-ua/2))$. Proposition 4, with $\mathrm{Var}N \le A \le \delta/(a\theta_0)$ and $\delta \le \delta_0 = (1+u)a$, gives

$$\frac{E_i}{\delta} \le 2 + \frac{8}{\ln 2}\sqrt{\kappa_0} + \varepsilon_0 \log_2\Big(e\Big[3 + \frac{8}{\varepsilon_0 a\theta_0}\Big]\Big) + \delta_0 \log_2\big(e[1+\delta_0^{-2}]\big).$$

Now let $L \ge 3$, so $a \le 1/8$ and $\sqrt{u} = 1/(8L) \le 1/24$. Then $\theta_0 \ge 1 - 1/1536$, $\varepsilon_0 < 0.157$, and $L\varepsilon_0 < 0.47$, since $La$, $L\sqrt{au} = 2^{-L/2}/8$ and $Lu$ decrease in $L$. Next $\kappa_0 < 1/(256L\cdot 0.999) < 0.00131$, so the second term is below $11.55\cdot 0.0362 < 0.42$. Since $\varepsilon_0 a\theta_0 \le 1$, the third term is at most

$$\varepsilon_0\Big[\log_2\frac{11e}{\theta_0} + L + \log_2\frac{1}{\varepsilon_0}\Big] < 0.157\cdot 5 + 0.47 + \frac{1}{e\ln 2} < 1.79.$$

Finally $\delta_0 < 0.126$, and since $x\log_2(e[1+x^{-2}])$ increases on $(0, 1/4]$ the last term is below $0.126\log_2(e\cdot 64) < 0.95$. Altogether $E_i/\delta < 2 + 0.42 + 1.79 + 0.95 < 6$. □

For terms of width at most three the bound $2 + \log_2(2e(2^w-1)) < 7.3$ of Section 2 holds at a variable of largest influence in the term, with no hypothesis on loads.

## 6. Bias

**Lemma 10.** *Let $f\colon \{0,1\}^n \to \{-1,1\}$, $m = \min(\Pr[f=1], \Pr[f=-1]) > 0$ and $\beta = \log_2(1/m)$. Then*

$$\frac{H(f)}{I(f)} \le 2 + \frac{2(n - 2 + \log_2 e)}{\beta}.$$

*Proof.* Let $v = 1 - \hat{f}(\emptyset)^2 = 4m(1-m)$ be the weight off the empty set. Conditioning on whether $U$ is empty, $H(f) \le -v\log_2 v - (1-v)\log_2(1-v) + v\log_2(2^n-1) \le v\log_2(e/v) + vn$, using $-(1-v)\log_2(1-v) \le v\log_2 e$. The right side increases in $v$ on $(0, 2^n]$ (its derivative is $n - \log_2 v$), and $v \le 4m \le 2$, so $H(f) \le 4m(n + \beta + \log_2 e - 2)$. A set $W$ of vertices of the $n$-cube spans at most $\frac{1}{2}|W|\log_2|W|$ edges [6, 8], so at least $|W|(n - \log_2|W|)$ edges leave it; with $W$ the minority set, $|W| = m2^n$, this gives $I(f) \ge 2\cdot 2^{-n}|W|(n-\log_2|W|) = 2m\beta$. Divide. □

For $\mathrm{AND}_n$ one has $m = 2^{-n}$, $\beta = n$, and both sides tend to 4. The bound is useful only when $m$ is exponentially small in $n$: if $\beta \ge n/10$, say, it gives $H/I < 22$.

## 7. Transversal designs

Let $Q$ be a prime power, $X \subseteq \mathbb{F}_Q$ with $|X| = w$, and let $\mathrm{TD}(w, Q)$ have variables $(\xi,\zeta) \in X \times \mathbb{F}_Q$ and terms $t_{\sigma,\tau} = \{(\xi, \sigma\xi+\tau) : \xi \in X\}$ for $\sigma,\tau \in \mathbb{F}_Q$. It has $n = wQ$ variables and $Q^2$ terms of width $w$; the point $(\xi,\zeta)$ lies on the $Q$ terms with $\tau = \zeta - \sigma\xi$; two distinct affine maps agree at most once, so the DNF is linear. Every variable has load $\lambda = Q2^{-(w-1)}$, and every variable of a term $t$ through $i$, other than $i$, has $\ell_j = (Q-1)2^{-(w-1)}$.

| $w$ | $Q$ | $n$ | terms | $\Pr[F=0]$ | $H/I$ | $\min_i E_i/\mathrm{Inf}_i$ | $\max_i E_i/\mathrm{Inf}_i$ |
|---|---|---|---|---|---|---|---|
| 3 | 3 | 9 | 9 | 0.4180 | 3.0856094956 | 2.1015004988 | 2.1015004988 |
| 3 | 5 | 15 | 25 | 0.1435 | 3.8797848238 | 2.7150952652 | 2.7150952652 |
| 4 | 5 | 20 | 25 | 0.3219 | 3.9024430124 | 2.4713311010 | 2.4713311010 |
| 3 | 7 | 21 | 49 | 0.0408 | 4.1345862211 | 2.5335124504 | 2.5335124504 |
| 5 | 5 | 25 | 25 | 0.5332 | 3.8799716886 | 2.2003373077 | 2.2003373077 |

TABLE 2. TD$(w, Q)$ with $X = \{0, \dots, w-1\}$, from exact Walsh transforms.

*Proof of Theorem 3.* Write $L = w - 1 = k + 2\lceil \log_2 k \rceil$, $a = 2^{-L}$, $\lambda = Qa = 2^{-2\lceil \log_2 k \rceil} \in [1/(4k^2), 1/k^2]$ and $\ell = (Q-1)a < \lambda$; note $w \le Q$ for $k \ge 3$ and $L/k \to 1$.

(1) $\lambda_j/a = Q$, far above $1 + 1/(64L^2)$.

(2) Fix $i$. It lies in $Q$ terms, all with $L_t = L$, so $\lambda_i = \lambda$, $\mu_t = L\ell$ and $\theta = 1 - L\ell - (Q-1)a/2 = 1 - O(1/k)$. Choose thresholds $k_t$. If at least $Q/2$ of them are at least 4, then $\sum_t \gamma_t \ge \frac{Q}{2} a \binom{L}{3} = \frac{\lambda}{2}\binom{L}{3} \ge \frac{L^3(1-o(1))}{48k^2} \to \infty$, so eventually $\varepsilon = 1$ and the third term of Proposition 6 is at least $\log_2(8e/\lambda) \to \infty$. Otherwise at least $Q/2$ of them lie in $\{1, 2, 3\}$, and a block with $k_t = r$ has $\alpha_t = \binom{L}{r} 2^{\binom{r}{2} - 2r} \ell^r / (1 - r\ell/2)$, so

$$\kappa \ge \frac{Q}{2} \cdot \frac{\min_{r \le 3} \alpha^{(r)}}{Qa\theta} = \frac{\min_{r \le 3} \alpha^{(r)}}{2a\theta}.$$

Since $\ell \ge \lambda/2$ and $1/a = Q/\lambda$, $\alpha^{(r)}/a$ is of order $\binom{L}{r}\lambda^{r-1}Q$, that is $LQ$, $Q$ and $Q/L$ for $r = 1, 2, 3$ up to bounded factors. So $\kappa \to \infty$.

(3) Let $Z$ be the number of satisfied terms and $\rho = 2^{-w}$. Then $\mathbb{E}Z = Q^2\rho \ge 2^k/(8k^2) \to \infty$. Terms with disjoint variables are independent; two intersecting terms have covariance $2^{-(2w-1)} - \rho^2 = \rho^2$, and each term meets $w(Q-1)$ others. Hence $\mathrm{Var}Z/(\mathbb{E}Z)^2 \le 1/\mathbb{E}Z + w/Q \to 0$, and $\Pr[F=0] = \Pr[Z=0] \to 0$ by Chebyshev's inequality. On the other hand the failures of the terms are decreasing events, so by Harris $\Pr[F=0] \ge (1-\rho)^{Q^2}$ and

$$\frac{\log_2(1/\Pr[F=0])}{n} \le \frac{Q}{w} \log_2 \frac{1}{1-\rho} = O\Big(\frac{2^{k-w}}{w}\Big) = O(k^{-3}).$$

Once $\Pr[F=0] < 1/2$ the minority value is $F = 0$, so $m = \Pr[F=0]$, and the bound of Lemma 10, which exceeds $2(n-1)/\log_2(1/\Pr[F=0])$, tends to infinity.

(4) Fix $i$ and let $N$ and $A$ be as in Section 4. Since $x_i$ is independent of $N$ and of the $A_t$, $\Pr[F=0] = A - \frac{1}{2}\delta$, and $\delta \le A\lambda$ by Lemma 7. Hence $\delta/\Pr[F=0] \le \lambda/(1-\lambda/2)$ and $\mathrm{Var}f/\delta = 4\Pr[F=0]\Pr[F=1]/\delta \ge 4\Pr[F=1](1-\lambda/2)/\lambda \to \infty$, uniformly in $i$, so the right side of Lemma 5 tends to infinity. □

For TD$(w, 2^k)$ the width bound of Section 2 is of order $w$, and Theorem 2 does not apply to any term. So a linear monotone DNF in which every restriction has a coordinate covered by one of these bounds satisfies $H \le C\,I$ by the induction of [18], but the designs show that not every linear monotone DNF has such a coordinate even at the root. Covering them would need either a separation estimate that uses the loads conditional on the background event $N$, rather than the unconditional loads, or a bound for averages over whole terms in the spirit of decision-tree arguments. Table 2 suggests that the actual ratios on the designs remain small.

## Acknowledgements and funding

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [24] and working together inside the Brain framework [26], a shared,

persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [25], the Math Brain repository [24] and the Brain framework repository [26].

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits programme (credit 530550103).

## Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

## Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, a project on the use of AI systems in research mathematics [26, 24]. A Codex agent running OpenAI's GPT-6.1 wrote the census programs, ran the census and found the certificates, the transversal-design obstruction and their proofs. Claude agents (Anthropic: Fable and Opus 5.5) checked the proofs, wrote the independent census check and the design computations, and prepared the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

## Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-linear-dnf-census`, archived on Zenodo with doi:10.5281/zenodo.24125383. The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [24].

## Appendix A. Notation

Symbols are listed roughly in the order in which they first appear, and the last column gives that place. The letters $a$ and $b$ are also the arguments of $\Phi$, $x$ and $y$ are real variables in the proof of Lemma 5, and $j$ and $z$ are the index and the variable of the logarithm series in Section 3.

| Symbol | Meaning | First appears |
|---|---|---|
| $f$, $n$ | a Boolean function $\{0,1\}^n \to \{-1,1\}$, and its number of variables | Section 1 |
| $x$, $x_j$ | a point of $\{0,1\}^n$, and its $j$-th coordinate; the $x_j$ are also the variables of a DNF | Section 1 |
| $i$, $j$ | coordinates; $i$ is the coordinate under study | Section 1 |
| $S$, $\hat{f}(S)$, $\chi_S$ | a set of coordinates, the Fourier coefficient of $f$ at $S$, and the character $(-1)^{\sum_{j\in S} x_j}$ | Section 1 |
| $U$ | the spectral sample, a random set with $\Pr[U = S] = \hat{f}(S)^2$ | Section 1 |
| $H(\cdot)$, $H(f)$ | Shannon entropy in bits, and the Fourier entropy $H(U)$ | Section 1 |

| Symbol | Meaning | First appears |
|---|---|---|
| $\mathrm{Inf}_i(f)$, $I(f)$ | the influence $\Pr[i \in U]$ of the coordinate $i$, and the total influence $\sum_i \mathrm{Inf}_i(f)$ | Section 1 |
| $C$ | the constant in the FEI conjecture $H(f) \le C\, I(f)$ | Section 1 |
| $E_i(f)$ | the erasure entropy $H(\mathbf{1}[i \in U] \mid U \setminus \{i\})$ | Section 1 |
| $\mathbf{1}[\cdot]$ | the indicator of an event | Section 1 |
| $c$ | an absolute constant with $E_i(f) \le c\,\mathrm{Inf}_i(f)$ at some coordinate | Section 1 |
| $F$ | a monotone DNF, with $f = 1 - 2F$; $F$ is linear if two of its terms share at most one variable | Section 1 |
| $t$, $\lvert t\rvert$, $w$ | a term of $F$, viewed as a set of variables, and its width; $w$ is a width | Section 1 |
| $\Delta$, $\nu$ | the largest number of terms through a variable, and the exponent in the hypothesis $\Delta \le 2^{(1-\nu)(w-1)}$ of [20] | Section 1 |
| $e$ | the base of the natural logarithm | Section 1 |
| $\mathrm{AND}_n$, $\mathrm{OR}_4$, $\mathrm{CYC}_4$ | the AND of $n$ variables, the OR of four variables, and the cycle DNF $x_1x_2 \vee x_2x_3 \vee x_3x_4 \vee x_4x_1$ | Section 1 |
| $\lambda_j$ | the load $\sum_{t \ni j} 2^{-(\lvert t\rvert-1)}$ of the variable $j$ | Section 1 |
| $L$ | the width of a term minus one; in Section 7, $L = w - 1$ | Theorem 2 |
| $m$ | $\min(\Pr[f = 1], \Pr[f = -1])$, the probability of the minority value | Section 1 |
| $Q$, $\mathbb{F}_Q$ | a prime power, and the field with $Q$ elements | Section 1 |
| $X$, $(\xi, \zeta)$ | a set of $w$ elements of $\mathbb{F}_Q$, and a variable of the design, a point of $X \times \mathbb{F}_Q$ | Section 1 |
| $\sigma$, $\tau$, $t_{\sigma,\tau}$ | the slope and the intercept of an affine map, and the term $\{(\xi, \sigma\xi + \tau) : \xi \in X\}$ | Section 1 |
| $\mathrm{TD}(w, Q)$ | the transversal-design DNF whose terms are the $t_{\sigma,\tau}$ | Section 1 |
| $k$, $F_k$ | the exponent in $Q = 2^k$, and the design $\mathrm{TD}(w, Q)$ with $w = k + 2\lceil \log_2 k \rceil + 1$ | Theorem 3 |
| $f^{(0)}$, $f^{(1)}$ | the restrictions of $f$ to $x_i = 0$ and to $x_i = 1$ | Section 2 |
| $g$, $h$ | $\frac{1}{2}(f^{(0)} + f^{(1)})$ and $\frac{1}{2}(f^{(0)} - f^{(1)})$; for monotone $F$, $h$ is the indicator that $i$ is pivotal | Section 2 |
| $\mathbb{E}$, Pr, Var | expectation, probability and variance over a uniform point $x$ | Section 2 |
| $T$ | a subset of $[n] \setminus \{i\}$ | Section 2 |
| $\delta$ | $\mathrm{Inf}_i(f) = \mathbb{E}h^2$ | Section 2 |
| $p(T)$, $q(T)$ | $\hat{g}(T)^2$ and $\hat{h}(T)^2$ | Section 2 |
| $\Phi(a, b)$, $\eta(x)$ | $\eta(a) + \eta(b) - \eta(a + b)$, and $-x \log_2 x$ | Section 2 |
| $N$, $M$ | the indicators that $F$ vanishes at $x_i = 0$ and at $x_i = 1$; $N$ is also the indicator that every term not through $i$ fails | Section 2 |
| $\Omega$, $\varepsilon$, $\kappa$ | a family of non-empty subsets of $[n] \setminus \{i\}$ and the two parameters of the separation inequality; their choice in Section 4 | Proposition 4 |
| $v$ | $\mathrm{Var} f$ | Lemma 5 |

| Symbol | Meaning | First appears |
|---|---|---|
| $\varphi(x)$ | $2x\ln 2-(x^2+1)\ln(x^2+1)+x^2\ln(x^2)$ | proof of Lemma 5 |
| $p_+$, $q_+$ | the sums of $p(T)$ and of $q(T)$ over non-empty $T$ | proof of Lemma 5 |
| $G$, $\mathrm{Aut}(G)$ | a linear hypergraph of the census, and its automorphism group | Section 3 |
| $K$ | an integer at most $4^n$ whose logarithm enters $H$, $I$ and $E_i$ | Section 3 |
| $B_t$, $L_t$, $a_t$ | $t\setminus\{i\}$ for a term $t$ through $i$, its size, and $2^{-L_t}$; $\lambda_i=\sum_t a_t$ | Section 4 |
| $t'$ | a second term through $i$ | Section 4 |
| $s$ | a background term, a term not containing $i$ | Section 4 |
| $\ell_j$ | the load of $j$ from background terms, $\sum_{s\ni j}2^{-(\lvert s\rvert-1)}$ | Section 4 |
| $\mu_t$, $\theta$ | $\sum_{j\in B_t}\ell_j$, and $1-\lambda_i^{-1}\left[\sum_t a_t\mu_t+\sum_{t'<t}a_{t'}a_t\right]$ | Section 4 |
| $k_t$, $r$ | a threshold $1\le k_t\le L_t$, and the number of variables a set has in $B_t$ | Section 4 |
| $\gamma_t$, $\alpha_t$ | the two quantities of the block $B_t$ in Proposition 6 | Section 4 |
| $J$ | a subset of $B_t$ with $k_t$ elements | Section 4 |
| $A$, $A_t$ | $\mathbb{E}N$, and the indicator that every variable of $B_t$ equals 1 | Section 4 |
| $R_t$, $V_t$ | the indicator that every background term disjoint from $B_t$ fails, and the OR of the conjunctions $s\setminus B_t$ over the background terms $s$ meeting $B_t$ | proof of Lemma 7 |
| $h_t$, $\Psi_t$ | $NA_t\prod_{t'<t}(1-A_{t'})$, and the factor in $h_t=A_t\Psi_t$ that does not depend on $B_t$ | proof of Lemma 8 |
| $D_J$ | the product over $j\in J$ of the derivatives $D_ju=\frac{1}{2}(u\vert_{x_j=0}-u\vert_{x_j=1})$ | proof of Lemma 9 |
| $R_J$, $Y_j$, $s_j$ | the indicator that every background term avoiding $J$ fails, the OR of the conjunctions $s\setminus\{j\}$ over the background terms $s\ni j$, and one such term | proof of Lemma 9 |
| $a$, $u$ | $2^{-L}$ and $1/(64L^2)$ | proof of Theorem 2 |
| $h'$ | $N(1-A_t)\mathbf{1}[\bigvee_{t'\neq t}A_{t'}]$ | proof of Theorem 2 |
| $\theta_0$, $\varepsilon_0$, $\kappa_0$, $\delta_0$ | the values of $\theta$, $\varepsilon$, $\kappa$ and an upper bound for $\delta$ in the one-term criterion | proof of Theorem 2 |
| $\beta$ | $\log_2(1/m)$ | Lemma 10 |
| $W$ | a set of vertices of the cube; in the proof, the minority set | proof of Lemma 10 |
| $\lambda$, $\ell$ | the load $Q2^{-(w-1)}$ of every variable of $\mathrm{TD}(w,Q)$, and the load $(Q-1)2^{-(w-1)}$ from background terms | Section 7 |
| $\alpha^{(r)}$ | the value of $\alpha_t$ for a block with $k_t=r$ | proof of Theorem 3 |
| $Z$, $\rho$ | the number of satisfied terms, and $2^{-w}$, the probability that a term is satisfied | proof of Theorem 3 |

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel